



CLINICAL RESEARCH EXCELLENCE STARTER GUIDE

*A Mini Guide to Clinical
Research Career Readiness*

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Intro to Clinical Research

What is Clinical Research? Clinical research is the process of studying medicines, treatments, medical devices, and other health care approaches in people. The goal is to gather reliable information about their safety and effectiveness.

What is a clinical trial?

A clinical trial is a structured research study involving human participants. Each trial follows a detailed research plan, called a protocol that describes who may participate, the procedures that will take place, the information collected, and how participants will be monitored throughout the trial. Participation in a trial is generally voluntary, and the informed-consent process explains the study, its procedures, foreseeable risks and potential benefits, alternatives, confidentiality, and participants' rights to discontinue participation.

What does work in clinical research mean & what will I do?

Clinical research is so much more than working with patients. Conducting a study responsibly takes a coordinated approach and a team of professionals. Depending on the role, a clinical research professional may be involved with:

- Identifying and communicating with potential study participants
- Coordinating study visits and procedures as outlined in the protocol
- Managing research documentation and data collection
- Supporting investigators and researchers
- Monitoring study conduct
- Working with sponsors and regulatory agencies
- Protecting subject rights, welfare, and safety

Clinical trials help to determine whether potential medical treatments are safe and effective. Clinical research professionals organize, conduct, document, and oversee those studies while protecting those who volunteer to participate.

Clinical Research Career Guide: Getting Your Foot in the Door

After gaining foundational experience, professionals can begin specializing:



To enter the industry, you don’t need a two- or four-year clinical or medical degree. Biopharma Academy of Clinical Research serves as a steppingstone for those looking to enter the field. Requirements do vary by employer and role.

CRC vs. CRA: What’s the Difference?

Clinical Research Coordinator (CRC) Overview: A Clinical Research Coordinator coordinates and facilitates clinical trial activities under the direction of the Principal Investigator. Their primary roles include the following:

- Participant Screening & Recruitment: Helping to identify potential study subjects and integrating screening and enrollment activities.
- Informed Consent Processes: Obtain and document the Informed Consent Process as outlined by GCP guidelines
- Study Visit Coordination: Schedule participants and coordinate study-related activities
- Protocol Compliance: Ensure study activities are conducted according to the approved protocol and applicable requirements
- Regulatory Documentation: Maintain essential documents and regulatory files

These are just a few notable takeaways for the CRC role. The CRC is one of the key people responsible for keeping a clinical trial organized at the research site.

Clinical Research Associate (CRA) Overview: A Clinical Research Associate or “friendly auditor” is primarily responsible for monitoring and overseeing the conduct and progress of clinical trials on behalf of the Sponsor.

- **Monitors Clinical Trial Sites:** Oversees how the study is being conducted at the site.
- **Reviews Study Data & Documentation:** Reviews study data and documentation for accuracy and adherence to the study protocol, including source data verification.
- **Conducts Site Visits:** Conducts Site Qualification, Site Initiation, Routine Monitoring, and Close-Out Visits.
- **Works with Investigators & Site Teams:** Collaborates with Investigators and research teams while maintaining appropriate trial documentation.
- **Ensures Protocol Compliance & Trial Integrity:** Monitors & verifies study information to support data quality, protocol adherence, and overall trial integrity.

The difference between the CRC and CRA: CRC helps conduct the trial at the research site, and the CRA monitors the site’s conduct of the trial on behalf of the Sponsor.

Salary Insights

Career List

- ✓ Clinical Research Nurse (CRN)
- ✓ Clinical Trial Assistant (CTA)
- ✓ Clinical Research Associate (CRA)
- ✓ Clinical Research Coordinator (CRC)
- ✓ Clinical Trial / Project Manager
- ✓ Quality Assurance Assistant
- ✓ Regulatory Assistant
- ✓ Business Development Assistant
- ✓ Finance Assistant

Earnings Potential

\$18-\$30/hr

Median Hourly Wage



Industry Acronym Basics

Learning clinical research is like learning a new language. In time, you will learn that there are many acronyms for words in the industry.

- **AE**-Adverse Event
- **CRA**-Clinical Research Associate
- **CRF**-Case Report Form
- **CTMS**-Clinical Trial Management System
- **eCRF**-Electronic Case Report Form
- **eTMF**-Electronic Master File
- **GCP**- Good Clinical Practice
- **ICF**-Informed Consent Form
- **IRB**-Institutional Review Board
- **NIA**-National Institute on Aging
- **PI**-Principal Investigator
- **Sub-I**-Sub-Investigator
- **CRF**-Code of Federal Regulations
- **CRC**-Clinical Research Coordinator
- **CRO**-Clinical Research Organization
- **DSMB**-Data Safety Monitoring Board
- **EDC**- Electronic Data Capture
- **FDA**-Food and Drug Administration
- **ICH**-International Council for Harmonisation
- **IND**-Investigational New Drug Application
- **NDA**-New Drug Application
- **NIH**-National Institutes of Health
- **SAE**-Serious Adverse Event

The NIA Glossary of Clinical Research Terms and NIH Commonly Used Abbreviations & Acronyms in Research provide reference definitions for many of these foundational terms.

Clinical Trial Workflow Guide

*There is a way a trial works.
Below is the proper process:*



1



Site Start-up

- Review the study protocol
- Evaluate whether the site has the patient population and resources to conduct the trial.
- Complete regulatory and essential study documentation
- Obtain required IRB/ethics approvals
- Prepare study systems and documentation
- Train the research team on the protocol and their responsibilities
- Participant Recruitment and Pre-Screening
- Identify potential participants who may meet the study's eligibility requirements

2



Participant Recruitment and Pre-Screening

- The site will begin to identify individuals who may meet the study's eligibility requirements.

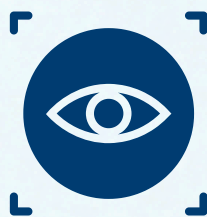
3



Informed Consent

- Ensure the Informed Consent Document contains all required elements according to the Code of Federal Regulations and in compliance with GCP guidelines.
- Understand that obtaining informed Consent is a PROCESS and an important component of subject protection and is subject to FDA requirements for clinical investigation.
- The Informed Consent Process must be adequately documented in each subject clinic record for compliance and review by all regulatory agencies.

4



Screening

- After consenting to the subject, protocol-required screening activities are performed to determine whether the individual meets the inclusion and exclusion criteria.
- If the subject fulfills all the eligibility criteria, the participant may continue in the trial.
- If the subject does not, then the participant is considered a Screen Failure.

5



Enrollment and Treatment/Intervention

- Study participants begin trial-related investigational treatment. This can be a device, intervention, or drug comparison as specified by the protocol.

6



Ongoing Subject Visits

- The protocol will specify the required procedures necessary for each study visit, and can include the following:
- Physical Exam
- Laboratory Assessments
- Height and Weight
- Investigational Product Review
- Diary Review
- Safety Evaluations

7



Monitoring and Quality Oversight

- Throughout the study, the sponsor's monitoring process provides oversight of the conduct of the study. A Clinical Research Associate (CRA) on behalf of the sponsor will come to the site to ensure the study is conducted in accordance with FDA and GCP guidelines.

8



Participant Completion

- Once the participant completes their final study visit, this does not mean the entire study is finished. When the Last study visits from the last enrolled participant from all the sites taking part in the trial has completed, then the trial has ended.

9



Post-Market Safety

- The FDA continues safety monitoring once all products are available for public use.
- Now that you know what clinical research is and the roles involved, let's dive deeper into what makes trials function and the processes behind them.
- Clinical trials are developed in phases. Think of these as different questions researchers are trying to answer as a treatment moves through development.



Understanding the Drug Development

From discovery to patients — a step-by-step journey of science, safety and hope.

01

Phase I

Is it safe?

(Safety + Dosage)

Phase I trials are “first-in-human” studies and follow the pre-clinical trials done in laboratory animals. Researchers study how the investigational treatment interacts with the human body, evaluate side effects, and gather dosing information.

02

Phase II

Does it work

(Efficacy + Safety)

During the second phase, researchers study the treatment in people with the disease or condition being investigated to gather preliminary effectiveness data while continuing to evaluate safety.

03

Phase III

Does it continue to work on a larger sample size

(Treatment Group)

These are larger trials used to confirm clinical effectiveness and continue evaluating side effects.

04

Phase IV

What happens after the treatment is approved?

(Post-Market Surveillance)

Phase IV occurs after the treatment has been approved by the Food and Drug Administration (FDA) and is still under surveillance to provide additional information about longer-term safety and effectiveness in broader populations

You can see a roadmap of most industry processes. Let's keep going into how a drug is developed.

1. **Discovery and Development- Could this be a medicine?**

Most often, development starts with an idea. Research begins in the lab with identifying and studying potential new drugs.

2. **Preclinical Research**

Laboratory and animal testing is performed to answer basic safety questions before clinical research in human subjects can begin.

3. **IND Application (Investigational New Drug)**

Before the investigational drug can move into clinical research, the sponsor gathers information supporting the initial human trials. The IND is the mechanism described by the FDA Investigational New Drug Application process for investigational drugs.

4. **Clinical Research Can Begin**

- Phase I- Safety + Dosage
- Phase II= Effectiveness + Dosage
- Phase III= Continued Effectiveness + Safety
- Phase IV=Post Marketing Surveillance

5. **New Drug Application (NDA)**

- The sponsor formally requests the FDA to review and approve the pharmaceutical for United States sale and marketing. Animal and human clinical trial data become part of the NDA.

6. **FDA Review**

- The regulatory agency team will examine the submitted information and determine whether to approve the drug/treatment.

7. **Approval**

- If the Food and Drug Administration determines that the evidence supports the drug's safety and effectiveness for its intended use, the drug may be approved.

8. **Post-Market Safety**

- The FDA continues to monitor all products once they are available for public use.

GCP—The Name of the Game

(Good Clinical Practice GCP)

GCP provides an ethical quality framework for clinical trials, and you'll hear these three words almost every day, multiple times a day during your career. Good Clinical Practice emphasizes subject protection, reliable trial results, risk-based quality management, clear investigator and sponsor responsibilities, and quality by design.

When you think of GCP, think of two BIG priorities:

- Protect the participant (safety, rights, and well-being)
- Protect the integrity of the trial (reliable data, reliable results, and overall quality)

Anytime you are about to perform any study-related task, ask yourself these questions:

- Are the subjects' rights, safety, and well-being protected?
- Am I following the protocol as it is outlined?
- Will the information contribute to reliable trial results?
- Do I understand my role and responsibilities?

Know the key players on the trial:

- **PI**=Principal Investigator
- **CRC**=Clinical Research Coordinator
- **CRA**=Clinical Research Associate
- **Sponsor**=Organization/person responsible for initiating and managing the trial
- **IRB**=Institutional Review Board

Complete the Missing Letter: Clinical Research

Can you complete these clinical research terms?

S _ O _ S _ R

P _ _ T _ C _ L

_ N V _ S T _ G _ T _ R

_ O N _ E _ T

R _ S _ A _ C _

_ A R T _ C _ P _ N T

_ L _ N _ C _ L

_ O N _ T _ R

S _ F _ _ Y

_ O C _ M _ N _ T

Answer Key

How many did you get right?

1. Sponsor

2. Protocol

3. Investigator

4. Consent

5. Research

6. Participant

7. Clinical

8. Monitor

9. Safety

10. Document

A Note of Thanks

Thank you for taking the time to explore this guide and strengthen your understanding of clinical research.

Whether you are taking your first steps into the field or building on your existing knowledge, we hope this resource has given you a clearer view of the clinical research journey, its key roles, and the principles that help advance clinical research.

Keep learning, keep asking questions, and keep building the knowledge and skills that can shape your career in clinical research.

Wishing you continued growth and success on your journey.



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